Smart Maintenance via Dynamic Fault Tree Analysis: A Case Study on Singapore MRT System

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***Abstract*—Urban railway systems, as the most heavily used systems in daily life, suffer from frequent service disruptions resulting millions of affected passengers and huge economic losses. Maintenance of the systems is done by maintaining individual devices in fixed cycles. It is time consuming, yet not effective. Thus, to reduce service failures through smart maintenance is becoming one of the top priorities of the system operators. In this paper, we propose a data driven approach that is to decide maintenance cycle based on estimating the mean time to failure of the system. There are two challenges: 1) as a cyber physical system, hardwares of cyber components (like signalling devices) fail more frequently than physical components (like power plants); 2) as a system of systems, functional dependency exists not only between components within a sub-system but also between different sub-systems, for example, a train relies on traction power system to operate. To meet the challenges, a Dynamic Fault Tree (DFT) based approach is adopted for the expressiveness of the modelling formalism and an efficient tool support by DFTCalc. Our case study shows interesting results that the Singapore Massive Rapid Train (MRT) system is likely to fail in 20 days from the full functioning status based on the manufacture data.**

Keywords–Dynamic Fault Tree Analysis; Smart Maintenance; Critical Infrastructure; Urban Railway System

I. INTRODUCTION

Singapore MRT is one of the heaviest urban railway systems in the world according to CoMET[1] (Community of Metros). The system features a daily ridership of nearly 3 million[2]. A disruption in the system could cause a huge effect on the service satisfaction. Unfortunately, due to the complexity of the system design and many legacy COTS (commercial off the shelf) products, which are error prone and unreliable, disrupted services happen frequently.

Maintenance of the system is done at the device level according to fixed maintenance cycles recommended by individual manufacturers. It is time consuming and less effective considering the variant failure frequencies of massive cyber and physical devices used in the system. However, it is natural to wonder whether, instead of focusing on maintaining individual devices, there is a smarter way to decide the maintenance cycle so as to reduce the frequency of the failures. That is to say, maintain the system/sub-systems rather than maintaining individual devices.

[1]http://cometandnova.org

[2]https://www.lta.gov.sg/content/dam/ltaweb/corp/PublicationsResearch/files/FactsandFigures/PT%20Ridership.pdf

In this work, we propose a data driven approach that calculates the mean time to failure (MTTF) value of the train service and the sub-systems using MTTF values of individual system devices. The results can be used to provide maintenance decision support for MRT operators. The challenges lie in modelling the dependability relations among devices and also between sub-systems. Thus, a DFT based approach is adopted for the expressiveness of the modelling formalism.

Fault tree analysis (FTA) [2] is a widely adopted technique to perform RAMS (reliability, availability, maintainability and safety) analysis in complex systems [18], [19]. DFT [3], [8] brings the concept of sequences into standard fault trees (sFT) since failures occurring in different orders may have very different effects on the system. For instance, the $PAND$ gate in DFT forces sequences to normal AND gate in sFT. Unless the failures happen from left to right, the gate will not output a failure signal. Although sFT is sufficient to model most features in a complex system, DFT advances in expressiveness and readability of the models. A detailed example is shown in Section II-A2. In addition, DFT analysis is well studied to perform two main tasks: 1) Qualitative analysis: finding the (minimal) cut set of low-level failures causing the top-level failure [14], [17]; 2) Quantitative analysis: computing the overall system reliability, mean time to failure of the system [4], [16] etc..

There are a number of practical experiences of applying FTA to railway systems. For example, the work in [11] and [10] applies FTA to a railway level crossing system and the work in [12] to a signal system in the high speed train. However, existing work usually studies a single sub-system for the most hazardous but rare happen events such as collision or derailment. Their model is considered less complex than ours where we model the full operating scenario of the train system which includes the train's on-board sub-system, signal control and communication sub-system and the traction power sub-system. In this work, we take the service failure as the top level event for analysis as it happens frequently and has no less importance in daily life.

Organization of the paper is as follows. Section II reviews the basics of DFT modelling and analysis. Section III describes the modelling of the three critical sub-systems, i.e., traction power sub-system, train communication sub-system and rolling stock sub-system, and the composition of the three models. Analysis of these models is presented in Section IV while feedbacks of industry partners on our results are listed as Questions and Answers in Section V. In the end, Section VI

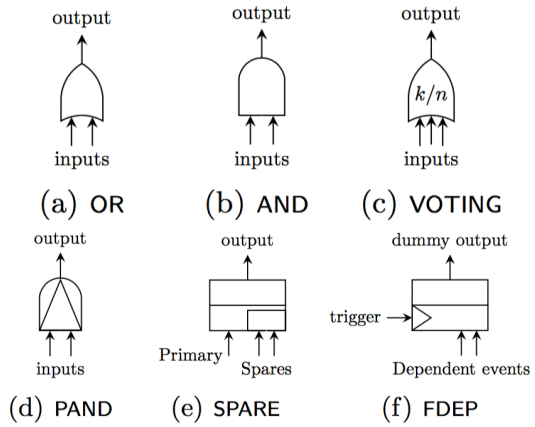


Fig. 1: DFT Gates

provides a review of the literature and conclusions are drawn in Section VII.

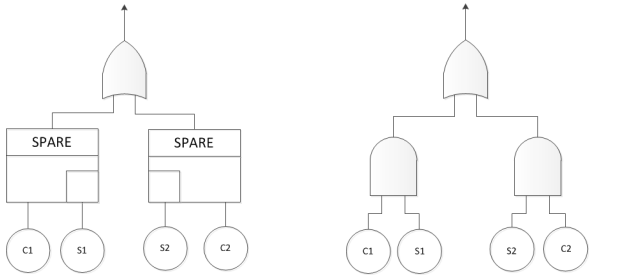
II. Preliminaries: Dynamic Fault Tree Analysis

A. Modelling

1) Basic Events, Gates: Dynamic fault trees (DFTs) is a graphic modelling technique that is often used in industry. It extends the standard fault trees (sFTs) with modelling constructs for dynamic behaviours such as sequenced failures. They are used to model how component failures propagates to the system failure. The leaves of a DFT are basic events (BE) which denote certain component failures while the root is labeled with a top-level event which is the target of analysis, a system failure. The non-leaves with gates express the dependence among multiple failures in the system.

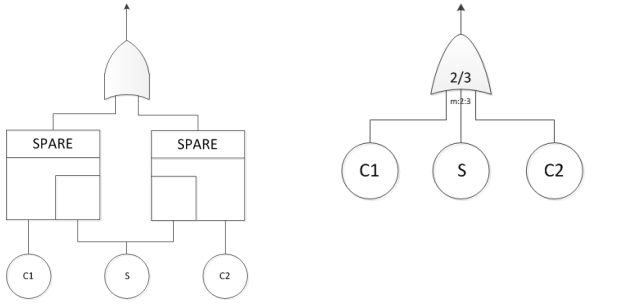
Gates consist of one or more inputs, and one output. Figure 1 depicts the DFT gates.

- The OR gate fails when at least one input fails.
- The AND gate fails when all of its inputs fail.
- The VOTING gate fails when at least k out of n inputs fail.
- The PAND gate fails when all of its inputs fail from left to right.
- The SPARE gate consists of a primary input and one or more spare inputs. At system start, the primary is active and the spares are in dormant mode. When the primary input fails, one of the spare inputs is activated and re- places the primary. If no more spares are available, the SPARE gate fails. Note that a spare component can be shared among several spare gates.
- The FDEP (functional dependency) gate consists of one trigger event and several dependent events. When the trigger event occurs, all dependent events fail. The FDEP has a "dummy" output, which is represented by a dotted line and ignored in calculations.



(a) DFT model of exclusive spare (b) sFT model of exclusive spare

Fig. 2: Models of Exclusive Spares



(a) DFT model of shared spare (b) sFT model of shared spare

Fig. 3: Models of the Shared Spare

2) DFT vs. sFT: Although in many cases, if proper abstraction is applied, sFT is sufficient for modelling complex systems, it often requires very careful thinking that a slight difference in the system would yield completely different sFT models. For example, consider a system composed of two critical components, $C1$ and $C2$, where if any one of them fails, the system will fail. In order to make the system more stable, spares are introduced for the components. If each of the components has its own spare, shown in the DFT model in Figure 2a, it is intuitive to model the system in sFT by replacing SPARE gates using AND gates as shown in Figure 2b. However, if a shared spare is used, as shown in Figure 3a, the same modelling strategy is not feasible, especially in the case that if $C1$ fails, spare S is switched on for $C1$ which implicates $C2$ has no spare any more. In this scenario, an equivalent model in sFT is using a voting gate as shown in Figure 3b. The resulting sFT model is not straightforward for users to figure out the dependability relations between $C1$, S, $C2$. Thus, for the purpose of easy comprehension of the models and easy communication with railway engineers, we adopted the DFT approach.

B. Minimal Cut Set

Minimal cut set (MCS) represents the minimal set of component failures that causes the failure of the whole system. Computation of MCS is utter important for identifying the bottlenecks of building a reliable system. For example, it can be used to find the single point failure in a system. In this work, we use a top-down algorithm [9] for computing it. The algorithm computes the cut sets of a fault tree starting from the

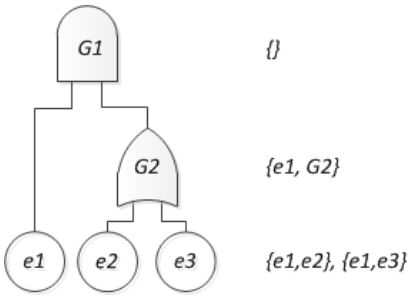
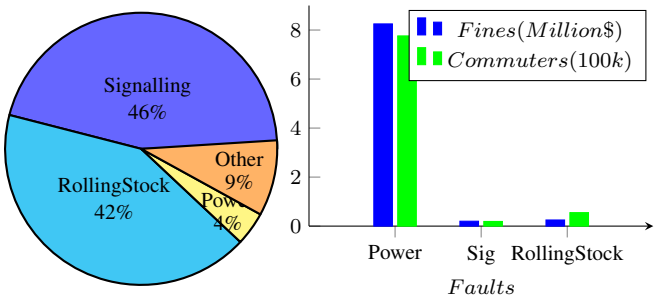


Fig. 4: A Top-down MCS Computing Algorithm



(a) Causes

(b) Fines and Affected Commuters

Fig. 5: Train Service Disruption Statistics

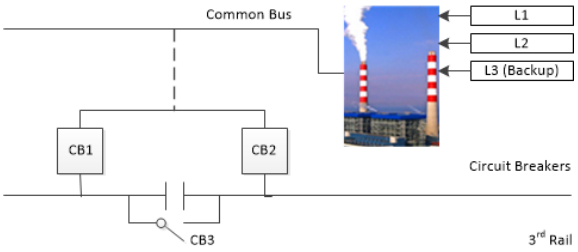


Fig. 6: Topology of Traction Power Sub-System

top event and expands these sets by computing cut sets of each gate encountered until basic events are reached. For example, as shown in Figure 4, for an AND gate $G1$ with basic event $e1$ and an OR gate $G2$ as inputs, where gate $G2$ takes $e2$ and $e3$ as inputs, event $e1$ and gate $G2$ both must occur to enable gate $G1$. The cut set starts as {e1, G2}. Furthermore, gate $G2$ will be enabled when either $e2$ or $e3$ occurs. Thus, the cut set further expanded as {e1,e2} and {e1,e3}. By removing the redundant elements, the minimal cut set is finalised as {e1,e2} and {e1,e3}.

C. DFTCalc Tool

DFTCalc is a powerful tool for modelling and analysing DFTs. The tool is based on continuous time models that each BE is equipped with a probability distribution f showing how the failure behaviour evolves over time, i.e. F(t) represents the probability that the BE is still running at time point t. The typical tasks of the tool are to compute the system reliability of a given DFT, i.e. the probability that the system has not failed within a given mission time T, the mean time to failure (MTTF), i.e. the expected time of a failure to occur, and the availability, i.e. the time that the system is up in the long. The deployment of statistical model checking (SMC) techniques as the computation back engine boost up the computation power of this tool. Interested readers are referred to [1].

III. System Description and DFT Modelling

A train service relies on the proper functioning of the rolling stock sub-system[3], the signalling sub-system residing in control stations and the traction power sub-system at trackside. Defining the 'train service failure' as an unexpected stop or major service delay during normal operation, any disruption in traction power sub-system, signal communication sub-system or rolling stock sub-system could cause it.

Besides, we did a survey about service disruption in Singapore's Circle Line MRT. Figure 5[4] summarises the top causes, fines and affected commuters of major incidents. It can be seen from the figure that major failures are sourced to faults in signalling system and rolling stock system. Typical causes are faults in the signal code generator which controls train speed, power trip etc.. Meanwhile, faults in the traction power system contributed to the longest delays which caused the most expensive fines.

Thus, in this work, we model three aforesaid important sub-systems, i.e., traction power system, signalling system and rolling stock system.

Meanwhile, for security concerns, in the following sections, we will skip confidential details about the system design but describe the interesting features of the system design.

Assumptions: In this work, we have certain settings and assumptions to clarify. Firstly, only manufacture data is used. The data mainly consists of MTTF of hardwares, provided by vendors. Operational statistics of hardware devices are interesting but not complete enough for our analysis. Secondly, only hardware devices used in both physical field and train-server communications are considered. Although we are aware that software fails more frequently than hardware, it is difficult to measure the life spans and failures of software components at the time of this work. Thirdly, in our model, we do not incorporate the effect of device aging and maintenance on the MTTF of hardwares. Fourthly, in practical, the operation of the train service consists of multiple instances of traction power sub-systems, train communication sub-systems and rolling stock sub-systems. However, due to the computation limit of existing tool, we only consider one instance of each. It is intuitive that the calculated MTTF of this train service model is the lower bound for that of the whole train system since all these instances will be composed by OR gates.

A. Traction Power Sub-System (TPSS)

TPSS is used for extracting power from power source to feed the train. The topology sketch is drafted in Figure 6.

[3]Rolling stock system refers to the train and its onboard train control system.

[4]Data presented in the following relies on the sources from SMRT Trains LTD (SMRT), Land Transport Authority of Singapore and Failrailsg.sg on http://failrailsg.appspot.com/index.html

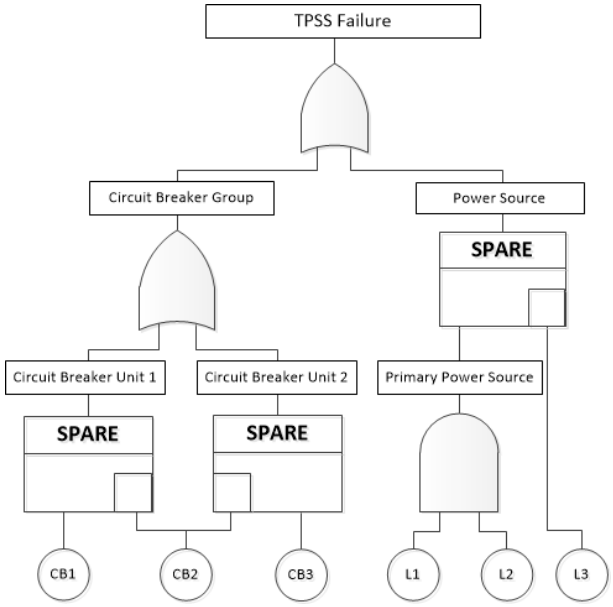


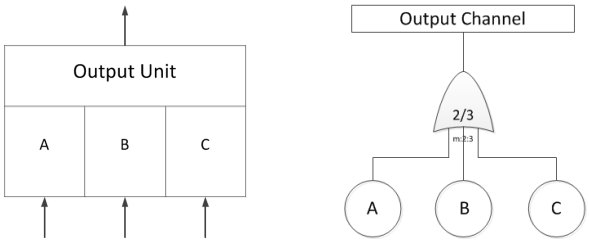
Fig. 7: DFT Model of Traction Power Sub-System

In TPSS, power is initially drawn from the public grid located in two locations with a backup power grid. Power is drawn to a common bus after being collected from the source, and then flows to each TPSS through certain devices to electrify the 3rd rail. Several circuit breakers are used to control the voltage and current in the system. Voltage and current are monitored at certain points. In the normal mode, circuit breakers are closed to transmit power to 3rd rail. These circuit breakers ensure current to flow throughout the 3rd rail despite the existence of neutral gaps. Power supply on 3rd rail can be switched off by opening the relevant circuit breakers. These circuit breakers on 3rd rail have redundancies. According to the design of neutral gap, three track feeder DC circuit breakers are used to control the power around one natural gap. Two more circuit breakers are regarded as backup for them. When the master circuit breakers fail, the backup can take over to control the power supply around the certain neutral gap.

Failure in power supply can be induced by several operational failures in TPSS such as the failure of public grid or failure of set of circuit breakers. The whole DFT consists of 4 OR gates, 4 SPARE gates, 2 AND gates and 14 basic events. A representative part of model is shown in Figure 7.

B. Train Communication Sub-System (TCSS)

Train communication sub-system is responsible for the communication between train and signalling server. In modern Communications-based train control (CBTC) systems, there are central automatic train supervision (ATS) servers monitor and control the train operations. A command from the ATS server traverses wireless and wired communication channels to reach the automatic train control (ATC) units resides both trackside and on trainborne. ATCs are used to control the train operations including accelerating, decelerating, braking etc.. At the same time, the train is reporting its status to ATS servers periodically.



(a) Topology (b) DFT Model

Fig. 8: A Device of Train Communication Sub-System

In order to ensure the data integrity, a consensus logic is designed in certain components. For example, there are three identical inputs from three running devices where the component will output the majority's data. Thus at least two of the inputs should be identical in order for the component to output a successful data, i.e., if faults occur in two input devices, the component fails. This features a voting gate as shown in Figure 8.

Failure in train communication system is induced by component failures, such as an actual defect of the communication media, a defect of trackside equipment, faults in wireless network, or failures in central server. The whole DFT of signalling system consists of 24 basic events connected over 5 SPARE gates, 1 VOTING gate, 2 AND gates and 7 OR gates.

C. Rolling Stock Sub-System (RSSS)

A rolling stock system is referred as the operation system deployed on trainborne. It draws power supplied by TPSS and is controlled by signalling system for operation. Thus, we separately describe this sub-system according to the power flow and the control flow. A simplified process is shown in Figure 9.

There are usually two motor cars on each head of a train. Each motor car consists of two bogies. To ensure normal operation, at most one bogie of the four bogies is allowed to fail. As shown in Figure 9, each bogie involves two shoes on each side of the train, PFD1, PFD2, PFD3 (PFD stands for power flow device which is used to replace the real names of these devices) and two traction motors. During operation, two shoes of a motor car on the same side, each from one bogie, are used for picking up power from 3rd rail. They are designed for redundancy purposes such that if one shoe fails, the train can still operate normally. However, if two shoes on the motor car are damaged at the same time, this car would not be able to draw any power from the 3rd rail. The train may still operate but in a degraded mode which results in reduced speed, or it may stop. The PFD1 and PFD2 are shared by two bogies and are used for transmitting electricity power. PFD3 receives control commands from signalling system to propel the two traction motors.

When a control message reaches the train, a trainborne ATC will issue commands to initiate the duty circle. This command is encoded and transmitted to CFDP (Control Flow Device- Propulsion). At the same time, CFDM (Control Flow Device- Motor) will be energised by ATC to decide if the

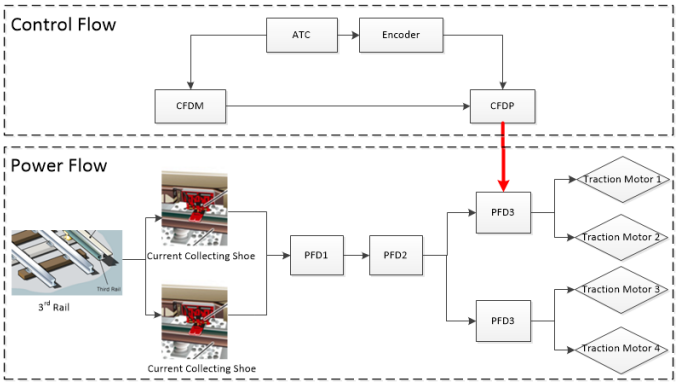


Fig. 9: Topology of Rolling Stock Sub-System

train is on propulsion mode and thus instruct CFDP to react accordingly. The control unit is shared by all four bogies thus we would consider the bogie is functionally dependant on the control unit, as shown in Figure 10. The rolling stock system model is then composed of four bogie models using a 2:4 voting gate.

Rolling stock failure is mainly sourced to propulsion failure which is induced by either disruption of power collection or the disruption of signal control. Propulsion signal control disruption will be induced from encoder's failure, trainborne ATC failure, CFDM failure or CFDP failure. Power collection disruption can be caused by a defect of current collecting shoe, a defect from propulsion unit, i.e. traction motor and PFD3, or a defect in breakers or switches. The whole DFT for propulsion scenario in rolling stock includes 34 basic events connected by 4 FEDP gate, 9 AND gates, 9 OR gates, and 1 VOTING gate.

D. Compose All Models

We composed the aforesaid three cases (i.e., TPSS, TCSS, and RSSS) into a whole DFT model, taking the train service disruption as a top event. Special interest is taken in the ATC components that are trainborne. The rolling stock sub-system is functionally dependent on them who belongs to signalling system. In this way, these two sub-systems are entangled. At the higher level, all the three sub-models are composed by an OR logic gate, as any of sub-system's failure would cause the whole system's failure, that is, train service disruption. The integrated DFT model for the whole system consists of 70 basic events connected over 9 SPARE gates, 4 FEDP gates, 2 VOTING gates, 13 AND gates, and 21 OR gates.

IV. Experiments and Results

Finally, two typical problems are being studied which are 1) is the system vulnerable to single point failures by computing the minimal cut sets? 2) what is the overall service reliability based on low-level devices' reliability information?

A. Minimal Cut Set Solution

1) Goal: Although redundancy design is widely adopted in the train system, the coverage is unknown. Thus, the purpose of this calculation is to find out if the system suffers single point failure. We use Dugan's top-down algorithm.

2) Results: Table I summarises the minimal cut sets. A complete computation over the whole system DFT model yields the result that there are 5 single point failures. Note that in the train communication system, there is only one media for wireless communication. Although this media is of high reliability, once it fails the whole system fails or in a greatly degraded mode. Besides, from our knowledge, the device PFD1_A is shared by 2 bogies on one motor car. Once it fails, two bogies in the motor car will fail causing the train system to be a degraded mode. It is similar in the case of PFD1_B on the other motor car, PFD2_A and PFD2_B.

TABLE I: Minimal Cut Set

Number	MCS
1	$wireless\ communication\ media$
2	$PFD1_A$
3	$PFD1_B$
4	$PFD2_A$
5	$PFD2_B$

B. Mean Time To Failure and Reliability Solution

1) Goal: To validate the composed DFT model, adopted an open source tool, DFTCalc introduced in Section II-C, to quantitatively analyse the train system. The web-version DFTCalc tool is used in this work, it takes the DFT models in Galileo [6] format. Besides, for safety and security purpose, anonymised data is used.

2) Results:

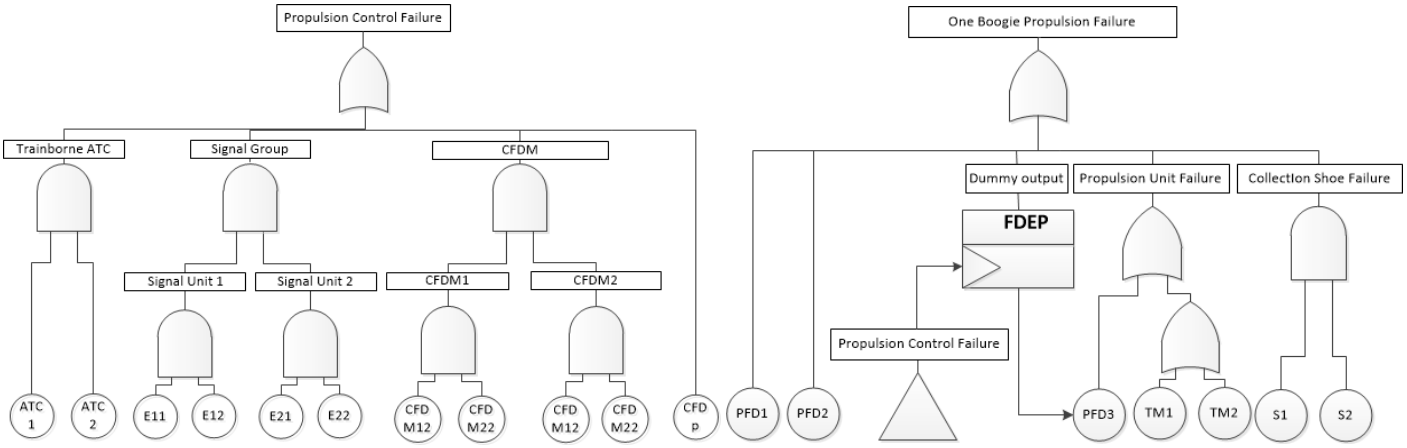
a) MTTF: For each of the DFT model, we plotted the mean time to failure of each sub-system and the probability that the system will fail over time. The results are shown in Figure 11 and Table II summarises the mean time to failure for each model.

TABLE II: MTTF and Computation Statistics

Model	MTTF (khrs)	#States	#Transition
TPSS	69.79 $\approx$ 8 years	240	1107
TCSS	0.4942 $\approx$ 21 days	580	3460
RSSS	4.877 $\approx$ 0.5 years	266	1154
TrainService	0.485 $\approx$ 20 days	116274	1144013

Note: Since the state space is too large for computation, we replace the TPSS system as a single event. The final result won't be affected since it is a standalone system, meaning no sharing components with other sub-systems.

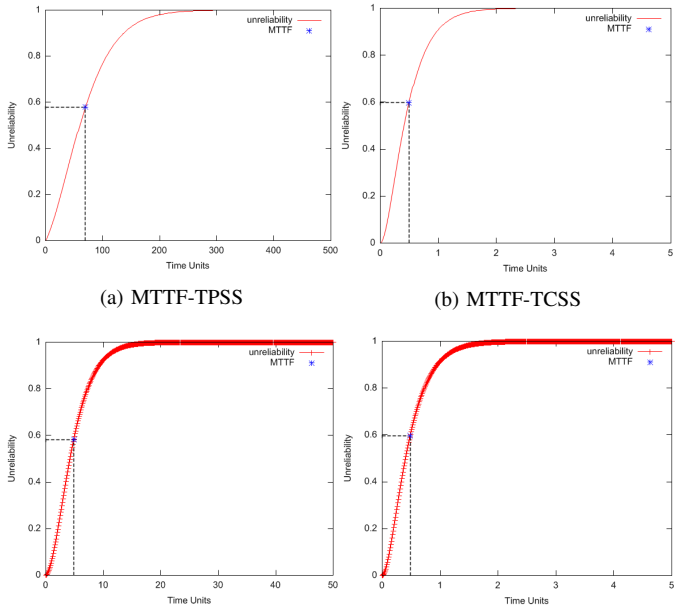
b) Failure Between Timed Interval: This experiment explores the distribution of failures along time, in other words, when is the failure most likely to happen. For an example, in the black curve of Figure 12, the second star point indicates that the probability of first system failure happens between 25k hours and 50k hours is highest. In the figure, the X axis stands for a timed interval and the Y axis stands for the probability of the first system failure. The legend indicates the time units used for each experimented sub-system. For example, $TPSS(10khrs)$ indicates that the time units used in black curve is 10,000 hours. The failure rate would first



(a) DFT Model of Propulsion Control

(b) DFT Model of One bogie

Fig. 10: DFT model of Rolling Stock Sub-System



(a) MTTF-TPSS

(b) MTTF-TCSS

(c) MTTF-RSSS

(d) MTTF-Train Service

Fig. 11: Mean Time To Failure (Time Units /khrs)

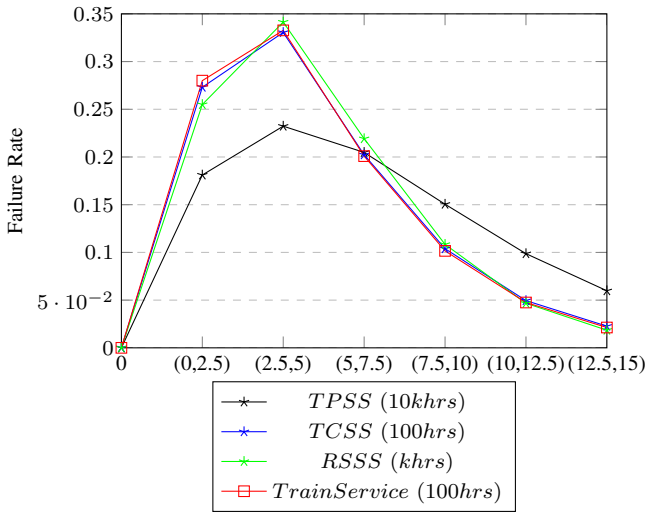


Fig. 12: Failure Rates Between (T1, T2)

increase and then drop, which suggests, it is less likely to fail in the very early stage of the running system and much more likely to fail when the system has run till the time around MTTF and afterwards, the unreliability will drop because most likely the system has already failed.

C. Observations

First, the introduction of redundancy designs, especially the hot standby mechanisms used in the MRT system greatly reduced the single point failures. However, there are still five cases where a single device's failure brings down the whole system. These cases might not be obvious to the operational management team when performing time consuming manual inspection. With our systematic analysis, industry people are able to thoroughly inspect the system for all possibilities. However, solutions to compensate for these single point failures are not known at the moment.

Secondly, from our experiment results, the train system is more vulnerable to failures in the signalling system. The MTTF value for the whole system model is very close to the signalling system's MTTF. Thus, we might conclude that the reliability of these cyber components is the bottleneck of building highly reliable train system. Relatively more effort should be done on this sub-system such as frequent thorough testing and maintaining of the communication system.

Thirdly, existing tools are incapable of handling practical industry systems even when we only model the core part of the system. In particular, the shared sub-trees make the model difficult for decomposed computation. Manual decomposition of the model is not a good option considering the complexity of the models. Recent development of utilising model checking

technique is a promising solution because there are many generic, efficient algorithms/ libraries and advanced state space reduction methods to be further developed.

V. Industry Partner Feedbacks

We've hold several rounds of meetings to explain our results to our industry partner from SMRT. They've given positive feedbacks and useful suggestion regarding the next steps of possible work.

A. Question A- Result Validation

Are these analysis results reasonably true from your observation of the system operation?

Answers: Personnel in charge of TCSS and RSSS replied in the affirmative base on their observation of the system operation. But the result for TPSS is hard to verify since the estimated MTTF (8 years) is longer than the Circle Line operation time which is 7 years.

B. Question B- Usefulness Feedback

Do you consider these analysis results useful? How would you think this analysis would help the train service operation and maintenance?

Answers: Original answer from TPSS staff-in-charge: "Yes, absolutely these analysis results are useful. With this, life expectancy of power equipment can be closely monitored and necessary steps can be taken such as PM (Preventative Maintenance) works, etc."

In practice, SMRT follows the maintenance cycle suggested by vendors. However, it is their vision to go into smart maintenance which is to replace/ repair the faulty equipment before it fails instead of performing maintenance in fixed time no matter there are failures or not. Thus, they think our result could be a useful reference in deciding the PM strategy.

C. Question C- Future Direction

Along this direction of analysis, what problem would be of interested to you and good for us to further explore?

Answers: Knowing that our analysis is based on manufacture data and hardwares, staff from train service operation management suggested the following areas for improvement.

- Environment factors e.g., human errors. For instance, operators may make wrong decisions that they allow the train to go without checking if the track is clear.
- Aging issues i.e., MTTF changes over time. Typically in TPSS and RSSS, some hardware failure is repaired on the spot that it is considered to have a shorter lifespan than original. Although it is unknown how MTTF changes over time for specific devices, there are operational data/ maintenance records which can be used for deriving the proper extrapolation.
- Software problems i.e., the equipments in TCSS encounter more problems caused by software than hardware. However, it remains a tricky issue to choose the right measurement for software failures.
- Repairs For example, a non-critical hardware failure is repaired before the system goes down. Thus, it should be studied how the repairs would affect the overall system failure probability?

VI. Related Work

There have been a number of methods proposed in the literature for qualitative or quantitative RAMS analysis of cyber physical systems.

A. FMEA Analysis

FMEA (Failure mode and effects analysis) [13] and its extensions with security, FMVEA (failure mode, vulnerability and effects analysis) [15], were the first analysis techniques which are still popular and used in the safety-critical industry including defence [7] and railroad [5] domains. These analyses offer a structured way to list possible failures, the consequences and countermeasures. Results from FM(V)EA can be useful for constructing fault trees, as they help in determining the possible component failures and thus the basic events.

B. Fault Tree based Analysis

The FFTA (formal FT analysis) approach [11] based on interactive proof construction is used to analyse a collision scenario at the level crossing of normal long-distance railroads.

This approach proved the functional correctness of the system and results in their paper showed that all minimal cut sets of the model were single point failures. The TFT (timed fault tree) [12] added delay to basic events and gates of the FTs. Its application to the signal system of a high speed train system enabled use of time as an attribute to evaluate the most critical part of the system, i.e. most urgent operations to take on system components, for example, calculating the minimum time between a fault and an accident.

In other works, DFT approach [10] was applied to analyse a major railroad crossing point in Netherlands. They adopted a tool based on statistical model checking to reason a traditional DFT model integrated with maintenance information, i.e., repair strategy. Their case study experiments showed that less possible repairs have marginal affects on the level crossing system.

Our work builds upon the above efforts with a tailored application to cyber physical systems (we haven't seen much effort in analysing CPS) where the features of shared cyber components complicate the analysis. Besides, as an industry system, the size of the model places a big challenge on existing tools, indicating that smarter algorithms should be designed to raise the computation at limits.

C. Applications of FTA to Railroad Industry

There are a number of attempts to apply FTA or its extensions to railroad systems. In [11], the authors used the Formal FTA (FFTA) approach to analyse a collision scenario at the level crossing of normal long-distance railroads. The FFTA approach is able to analyse functional correctness with interactive formal proof construction. The result shows that all minimum cut sets of the model is a single point failure.

In [12], the authors used Timed Fault Trees (TFT), which extended the SFTA with time labeled at each basic event as well as the gates, to estimate the delays. They've applied TFT to the signal system of a high speed train system. They are able to use time as a metric for evaluating the most critical part of the system, i.e. the most urgent operations to be undertaken on system components. For example, calculate the minimum time between a fault and an accident. Their result is also able to help railway operators set the standard of maintenance.

In [10], the authors used DFTA to analyse a major railroad crossing point in Netherlands. They've used a tool based on statistical model checking to reason a traditional DFT model integrated with maintenance information, i.e., repair strategy. Their case study experiments showed the result that less possible repairs have marginal affects on the level crossing system.

VII. Conclusion

This work presented a data driven approach based on DFT analysis to study the failures in the urban railway system, taking Singapore's MRT system as an example. The results of this work can be used to analyse the reliability of train system and provide support for deciding smart maintenance strategies, so that relatively more efforts can be spent on maintaining the operation of critical subsystems rather than individual devices. In the work, we modelled the complex system features such as the shared spares and functional dependency between sub-models. The analysis results demonstrated that though the system was less vulnerable to single point failures, the train service was more likely to fail due to the low reliability in the train communication sub-system. Thus, thorough testing and maintenance of the communication system should be done more frequently than that of other sub-systems. Feedback from our industry partners is encouraging about the usefulness of our approach and its applicability to practical settings. Our work could be generic to most cyber physical systems, though effective analysis tools are still insufficient. For example, the complete system model had a state space estimated to be 10^9 and the tool DFTCalc was not able to handle it.

This work also reveals more interesting problems to investigate in future, especially in the aspect of 1) replacing the manufacture data with operational data such as mean time between failure (MTBF) to improve the usability of the results; 2) comparing maintenance schemes and their consequent improvements in reliability of the system.

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